



Database Management Systems

UE24CS351A -Introduction

Anchor: Dr. Nagasundari S, Dept of CSE

Database Management Systems

Unit 1: Introduction to Database Management and SQL

Slides adapted from Author Slides of “Database System Concepts”, Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition, 2019. And Author Slides of Fundamentals of Database Systems”, Ramez Elamsri, Shamkant B Navathe, Pearson, 7th Edition, 2017.

Special Acknowledgement to all CSE Professors who contributed to this endeavor

Dr. Nagasundari S

Department of Computer Science and Engineering





Emergency Exit



Assembly Point



Washroom



No Chatting



Phones on silent



No Sleeping

- This presentation is purely educational.
- The views expressed by the presenter is not representation of any organization.
- The views are based on professional experience of the presenter and no liability is accepted by the presenter in the event of any potential or perceived losses resulting from this presentation.

Course Objectives:

- To introduce fundamental concepts of database systems, relational algebra, and basic SQL querying.
- To develop the ability to use advanced SQL techniques and understand the strategies for efficient query processing and optimization.
- To equip students with normalization principles, functional dependencies, and transaction management to ensure data integrity.
- To understand the modern data management systems including NoSQL - graph and vector databases.

At the end of this course, the student will be able to:

- Demonstrate an ability to explain the structure of a database system, ER diagrams, relational algebra, and write basic SQL queries involving constraints and keys
- Construct and optimize SQL queries including nested subqueries, joins, views, triggers, and stored procedures
- Apply normalization techniques using functional dependencies and design schemas, and ensure the implementation of transaction and concurrency control mechanisms.
- Create applications of graph and key value databases, and develop basic application using vector database

Unit 1: Introduction to Database Management

Purpose, View of data, Database design- Introduction to databases, Database application architecture, E-R Model, reducing ER to a relational schema. Relational operations (Algebra), Unary Operations - Unity, Binary, Aggregate Functions, Grouping, SQL overview, Data definition, Structure of SQL queries, constraints and keys. **14 Hours**

Unit 2: Relational Model and Database Design

Additional Basic Operations, Set Operations, Null Values, Aggregate Functions, Nested Subqueries, Database Modification, Join expressions, Views, Triggers, Functions, and Procedures, CTE, windows functions and full text search functions. **14 Hours**

Unit 3: Advanced Design Concepts and Implementation

Introduction to strategies of Query processing and Query optimization, Functional Dependencies, Inference Rules, Normal Forms Based on Primary Keys (1NF, 2NF, and 3NF), General Definitions of Second and Third Normal Forms Boyce-Codd Normal Form, Higher Normal Forms.

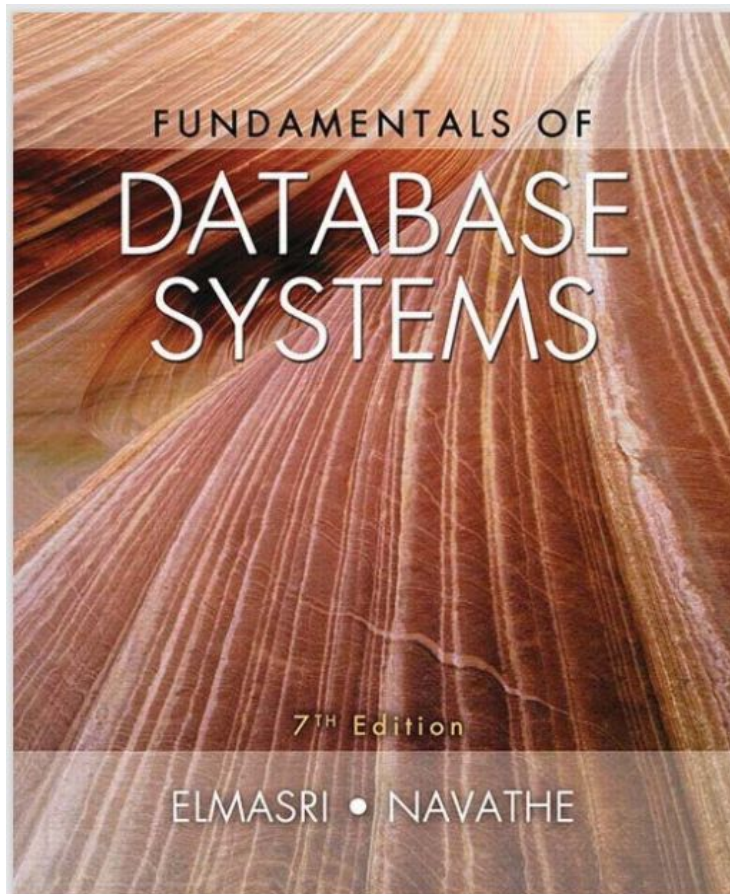
14 Hours

Unit 4: Advanced Databases

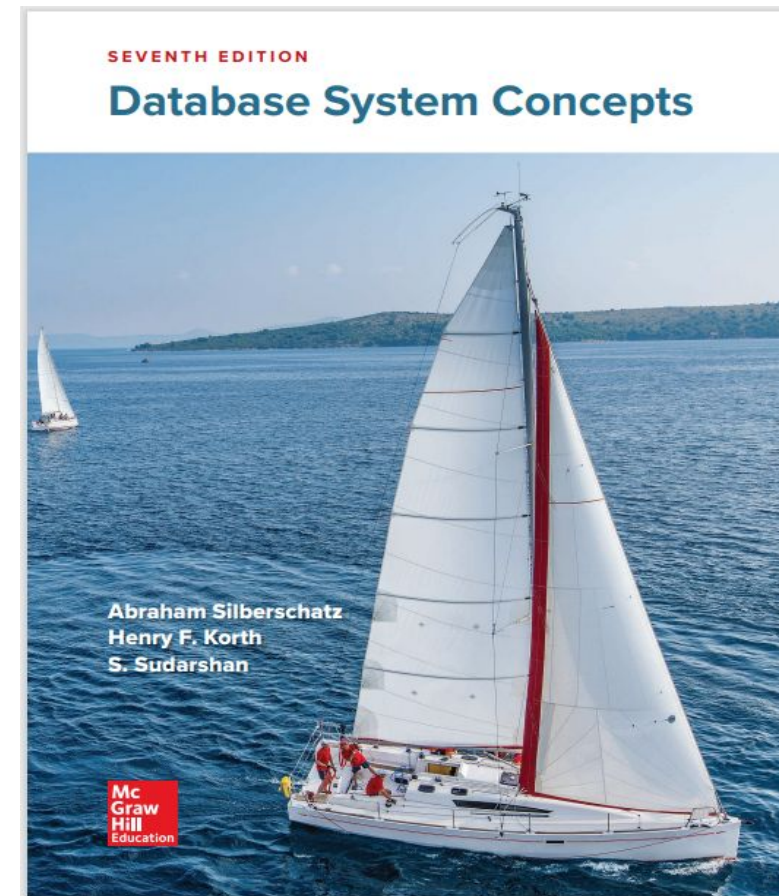
Database transactions, Concurrency control, Locking, Introduction to NoSQL databases, Key-value database, Graph databases (Neo4j), Introduction to Vector databases, Vector search with Relational and NOSQL databases. Web Application development using Next.js and SQL

14 Hours

Textbook 1 (TB1)



Textbook 2 (TB2)



Text Book(s):

1. “Fundamentals of Database Systems”, Ramez Elmasri, Shamkant B Navathe, Pearson, 7th Edition, 2017.
2. “Database System Concepts”, Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition.
3. Essential Concepts of Vector Databases: Mastering Vector Databases for Advanced Data Management by Paulo Dichone, 1st edition, Packt publisher

Reference Book(s):

1. Database Management Systems, R Ramakrishnan, J Gehrke, 3rd Edition, McGraw Hill, 2002.
2. Data on the Web: From Relations to Semistructured Data and XML, S Abiteboul, P Buneman, D Suci, Morgan Kaufman, 1999.
3. “An Introduction to Database Systems” by C. J. Date, Pearson Education, 8th Edition, 2003

Theory Components	Conduction Marks	Scaled Marks
ISA- 1	40	20
ISA- 2	40	20
Level 2 Experiential learning Quiz	20 to 30	10
ESA	100	50
Total		100

Experiential Learning Components	Conduction Marks	Scaled Marks
Level 1 Experiential Learning Exercises	7 Exercises * 10 Marks =70 Marks	10
Level 3 Experiential Learning Project	Demo- 7 marks Modification and Viva- 3 marks	10
Total (FSA Component)		20

Level 1 Experiential Learning Sessions

1. Design of ER Model
2. ER Model to Relational Schema and DDL commands Execution
3. DML and Join SQL queries
4. Nested and Correlated SQL queries
5. Procedures and functions
6. Triggers and Isolation level in transaction management
7. Redis CRUD Operations
8. Neo4J (Cypher query language)

Tools/ Languages: MySQL Workbench, Python, Redis, Neo4J, ERwin or any draw tool for ER modeling

Level 2 - Experiential Learning

- Test and evaluate knowledge of higher-order SQL ,NoSQL and Vector database queries through a comprehensive quiz.

Level 3 - Experiential Learning

- Develop a modern database application that combines SQL, NoSQL, and Vector Databases to efficiently store, manage, and retrieve different types of data.
- The application will use Next.js to provide
 - a responsive web interface,
 - enabling users to interact with the databases and
 - perform intelligent data searches.

Range – Marks ¹	Grade Letter
90 to 100	S
80 to 89	A
70 to 79	B
60 to 69	C
50 to 59	D
40 to 49	E
Below 40	F (Fail)

¹Range May be adjusted as appropriate

- If any assignment or lab work is submitted late, the following applies:
 - Late by up to 24 hours (1 day) – lose 25% of your marks
 - Late by up to 48 hours (2 days) – lose 50% of your marks
 - Late by up to 72 hours (3 days) – lose 75% of your marks

If you have accentuating circumstances, please let us know earliest and positively before the deadlines; we will try to accommodate.

Understanding Plagiarism

Mandatory Exercise

Department of Computer Science and Engineering



- Plagiarism is defined as presenting someone else's work, including the work of other students, as one's own.
Any ideas or materials taken from another source for either written or oral use must be fully acknowledged, unless the information is common knowledge.
What is considered "common knowledge" may differ from course to course.
- A student must **not** adopt or reproduce ideas, opinions, theories, formulas, graphics, or pictures of another person **without** acknowledgment.

- A student must give credit to the originality of others and acknowledge an indebtedness whenever:
 - Directly quoting another person's actual words, whether oral or written;
 - Using another person's ideas, opinions, or theories;
 - Paraphrasing the words, ideas, opinions, or theories of others, whether oral or written;
 - Borrowing facts, statistics, or illustrative material; or
 - Offering materials assembled or collected by others in the form of projects or collections without acknowledgment

- For those NOT familiar, please visit the links and follow the instructions.
 - [University College London- Learn about Plagiarism](#)
 - [Welcome to the Indiana University Plagiarism Tutorials and Tests - Learn how to recognize plagiarism, test your understanding, and earn a certificate.](#)
 - Take the certificate for Master's level.
- Any plagiarism will **result in Zero marks** for the assignment/work
- And in addition, **downgrade** of final subject result **by ONE grade!**
- For **both students giving** and students **taking** e.g. B to C, D to E

Honour code, Ethics and Academic Honesty

Department of Computer Science and Engineering



- **Asking questions is encouraged**
 - Discussing course topics with others is welcome
 - Peer Learning is Encouraged
- **Limits of collaboration**
 - Don't share your Homework, Assignment and Lab solutions with anyone else; And Vice versa;
 - You should never see or have possession of anyone else's solutions, including from past semesters
 - Dishonesty will result in severe penalties

- The Honour Code is PESU statement on academic integrity.
- It articulates our expectations of students and faculty in establishing and maintaining the highest standards in academic work.
- The Honour Code is an undertaking of the students, individually and collectively:
 - *Students will not give or receive aid in examinations and class work, labs and assignments;*
 - *Students will not give or receive unpermitted aid in class work, in the preparation of reports, or in any other work that is to be used by the instructor as the basis of grading;*
 - *Students will do their share and take an active part in seeing to it that others as well as themselves uphold the spirit and letter of the Honour Code.*

- Academic honesty is **expected** from each student participating in the course.
- **No sharing** (willing, unwilling, knowing, unknowing) of assignment code or solutions between students;
- Labs and Assignments must be done by each student independently.
- **No submission** of downloaded code (from the Internet, Campus LAN, or anywhere else) is accepted.
- Any dis-honesty will result in will result in **Zero marks** for the assignment/work

And in addition, **downgrade** of final subject result **by ONE grade!**

This applies for **both** students giving and students taking e.g. B grade will downgrade to C

- In working on assignments, you can use external references
 - Websites
 - Academic papers
 - Generative AI systems (e.g., ChatGPT)
- But you must provide citations to all external resources that you use
 - URL of websites
 - Citation of academic papers, ideally with URL of PDF
 - For Generative AI systems, provide all prompts that you used as you worked on your solution
- Ultimate educational goal is for **you to understand** the course's core contents; please use any references that support your learning
- If a reference (e.g., a Generative AI system) supports your completion of an assignment but **not your learning**, please do not use it in such a way

- ChatGPT and related tools are useful, but have risks:
 - Providing incorrect information, simply copying text from other sources, undermining learning goals of an assignment in which we are asking you to think critically.
- In this class:
 - Whether you use generative AI or read someone's blog post or code, you are expected to ensure that the **final product is your own, original work, which you should be able to explain.**
 - You may use generative AI unless specified otherwise for an assignment, but you must **reference what tool you used, as well as provide the prompts you used.**
 - You are responsible for ensuring that the resulting text is **correct**, that any included ideas are properly cited, and that there is no text plagiarized from other sources.
 - Please note that using such tools may imply donating your data to companies.
- **Finally, note that if the results you submit are narrower and shallower in scope this may impact your grade.**

- Most of all: Eagerness to learn!
 - We expect you to push yourself to learn as much as possible.
 - We expect you to be a strong, independent learner capable of learning new concepts from the lectures, the readings, and on your own.
 - It is okay and expected if you are not able to get every possible point.
- At the same time: Take care of yourselves and communicate with us!

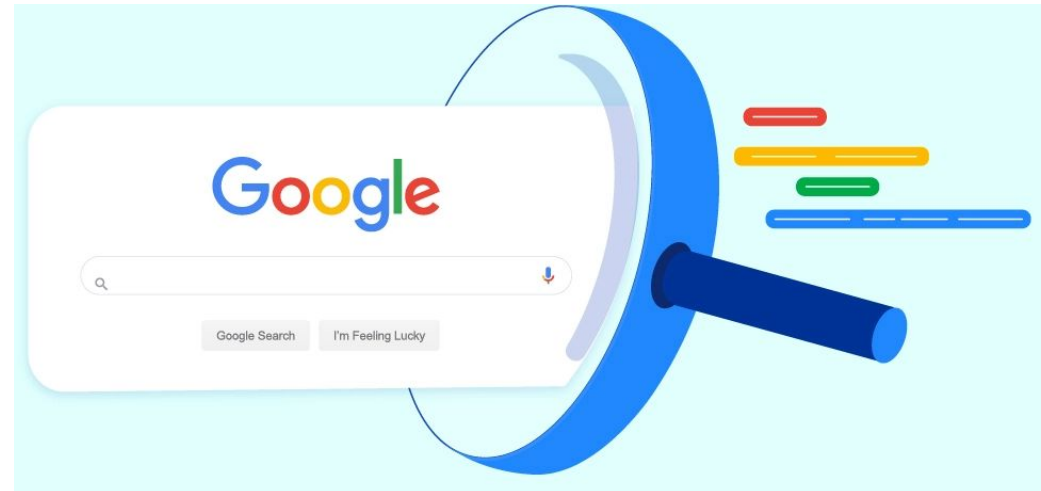
- Everyone in this class **deserves** to be in this class!
- We are all coming to this course with **different backgrounds** and experiences
- **Ask questions; be participative**
- There are **no bad questions**; we are always supportive
- We aren't always aware of everything, so **please call our attention to things as needed**

- In this course, you will be exposed to accessing data from databases.
- To be clear, **you are not to use this or any other similar information to test the security of, break into, compromise, or otherwise attack, any system or network** without the express consent of the owner.
- In particular, **you will comply with all my instructions when doing the labs.**
 - My instructions are in consonance with applicable laws of India and PES University policies.
 - If in any doubt, please consult your professor!
- Any violation is at **YOUR RISK!**
- **And may result in severe consequences.**

- Have you ever wondered on how a large e-commerce website like Amazon is able to retrieve the products that you would want to purchase?
- Nowadays social media (eg: Instagram) one of the most active platforms where people post their day to day activities etc. how and where is all these information stored? By typing a person's name we can get to see their entire profile, posts that person has posted how is this possible?
- In a bank, multiple transactions take place everyday. Even if 2 transactions are happening simultaneously from account, still the money in the accounts is properly deducted and credited how does this happen?



- Suppose you visit your doctor for a general checkup. Now by typing in your name or your ID, all your medical history details would be visible to the doctor. How does this happen?
- When you search for a specific topic on Google, how does the search engine quickly retrieve relevant web pages from millions of websites and present them to you?

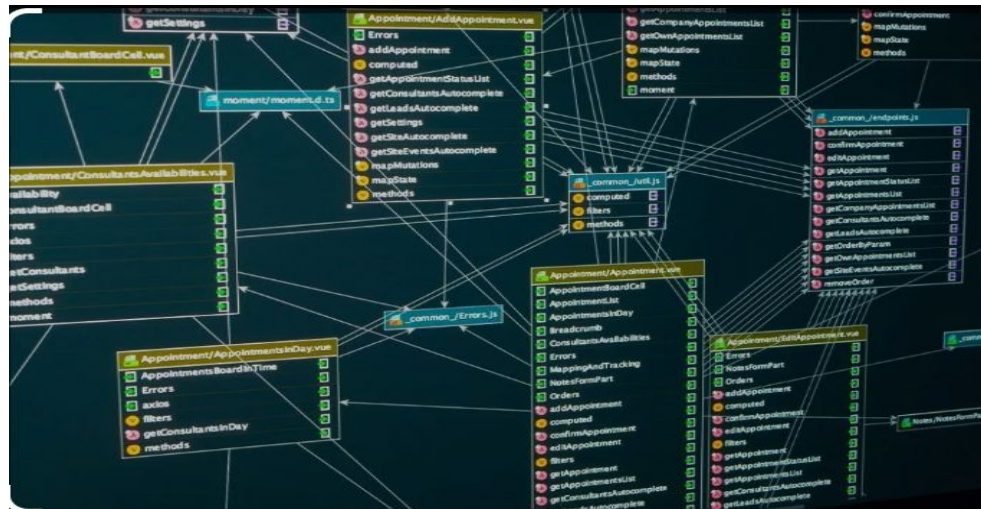


All the above questions have a single answer and that is **DATABASE**

Before going ahead and understanding what database exactly and how is designed and used for all the other application, let us try answering a few more questions:

- What do you mean by **Data**?
- How is it different from **Information**?

In current world, there is a lot of data that is available, how are all these stored or organized so that we can get the required information?

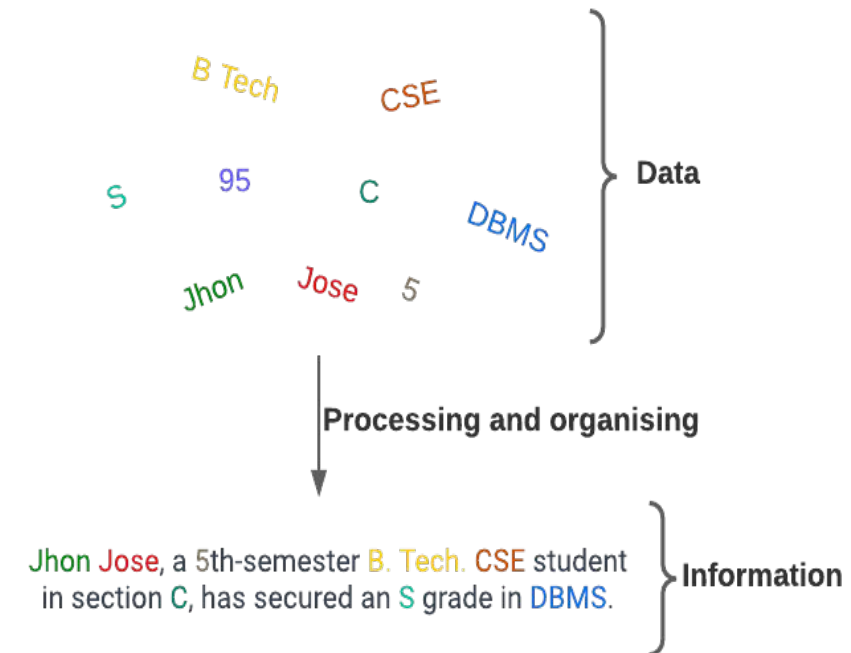


Data

- Data represents the raw material, like individual puzzle pieces, that serves as the foundation for knowledge.
- Data is a raw and unorganized fact that is required to be processed to make it meaningful.

Information

- When data is processed, organized, structured, or presented in a given context to make it useful is called information.
- Data is a collection of facts, while information puts those facts into context
- It is organized and is utilized by humans in some significant way to make decisions and draw some conclusions



**Data doesn't depend on
information, Information depends
on data**

Database

- A database is a **collection of related data** representing some aspect of the real world, also called the **mini-world** or the **universe of discourse (UoD)**.
- It is a **logically coherent collection**, meaning it is not just a random assortment of data but is organized with inherent meaning and structure.
- Databases are designed, built, and populated for a **specific purpose**, catering to the needs of applications or systems that interact with them.
- They can **vary in size and complexity**, ranging from small databases used by individual applications to large-scale enterprise databases handling vast amounts of data.

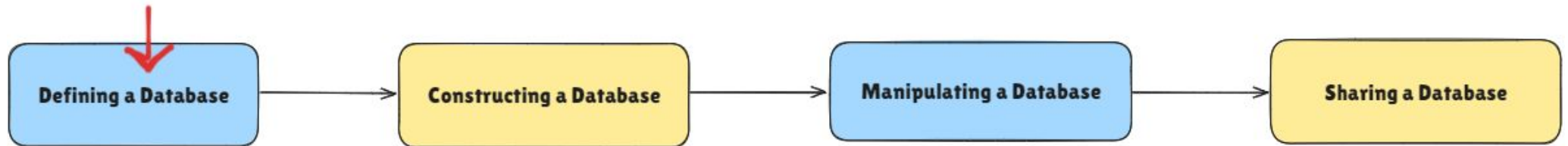
Could you think of what do these individual terms mean w.r.t student information Database?

Database Management System

- A modern database system is a complex software system whose task is to manage a large, complex collection of data.
- DBMS contains information about a particular enterprise
 - Collection of interrelated data
 - Set of programs to access the data
 - An environment that is both convenient and efficient to use
- Database systems are used to manage collections of data that are:
 - Highly valuable
 - Relatively large
 - Accessed by multiple users and applications, often at the same time.

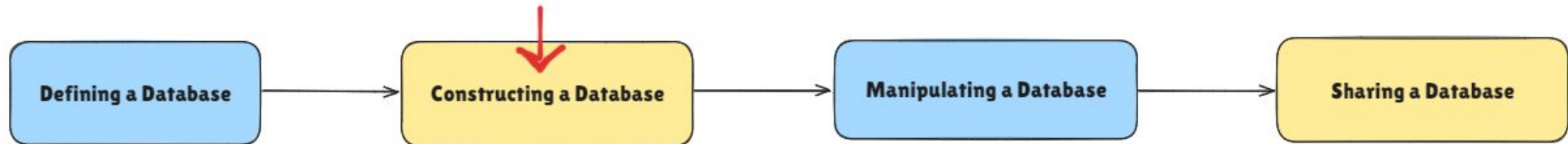
Let us break these statements down further and understand them in the context of the student information database

Database Management System



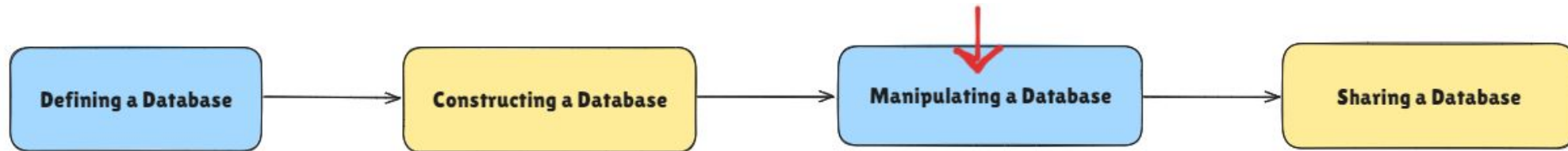
- A general-purpose software system that facilitates the processes of defining, constructing, manipulating, and sharing databases among various users and applications
 - **Defining a database** involves specifying the data types, structures, and constraints of the data to be stored in the database. By defining the structure and data types, the DBMS ensures the consistency and integrity of the stored data.
 - The "Students" table may include fields such as "Student ID" (numeric), "Name" (text), "Email" (text), "Date of Birth" (date), and "Major" (text).
 - The "Courses" table may have fields like "Course ID" (numeric), "Course Name" (text), "Credits" (numeric), and "Instructor" (text).

Database Management System



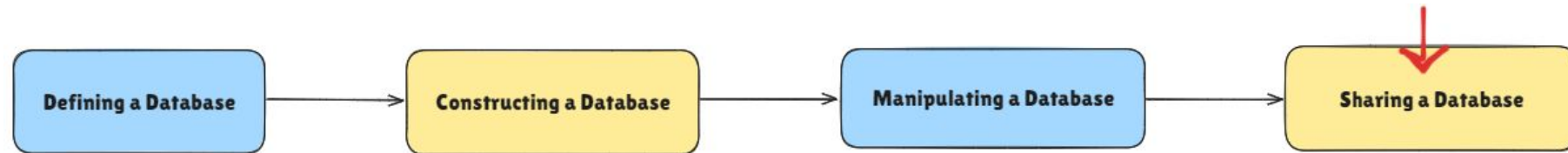
- A general-purpose software system that facilitates the processes of defining, constructing, manipulating, and sharing databases among various users and applications
 - **Constructing the database** is the process of storing the data on some storage medium that is controlled by the DBMS. It organizes the data in a way that allows for efficient retrieval and manipulation. The DBMS ensures data is securely stored and can be accessed by authorized users or applications.
 - The DBMS may store the Student Information Database on a server with appropriate data storage capacity and performance capabilities.

Database Management System



- A general-purpose software system that facilitates the processes of defining, constructing, manipulating, and sharing databases among various users and applications
 - **Manipulating a database** includes functions such as querying the database to retrieve specific data, updating the database to reflect changes in the mini world, and generating reports from the data. It ensures that any changes made to the database follow predefined constraints and maintain data consistency.
 - A university administrator may use SQL queries to retrieve specific information, such as "Retrieve all students enrolled in Computer Science courses." The DBMS processes the query, retrieves the relevant data from the database, and presents it to the user.
 - Allows authorized users to update the database, such as adding new students or updating grades.

Database Management System



- A general-purpose software system that facilitates the processes of defining, constructing, manipulating, and sharing databases among various users and applications
 - **Sharing a database** allows multiple users and programs to access the database simultaneously, provided they have the necessary permissions
 - Faculty members, administrative staff, and students can access relevant information concurrently, as long as they have the necessary permissions.
 - For example, faculty members can access student details for their courses, while administrators can update and maintain the overall database.

Some questions to ponder upon

- 1) How do you define data, and how does it differ from information in the context of database management systems?
- 2) Explain the role of a DBMS in managing highly valuable and relatively large collections of data. Why is this important for organizations?

Databases help us to:

- Store large amounts of data (file structure, disk management)
- Understand the data (data models)
- Keeps data secure (security, recovery)
- Find required data and use/manipulate it (query languages, concurrency control, and data analysis tools)
- Get accurate information (as databases have built-in constraints and checks help in this)
- Maintain Data integrity (ensures data is accurate and consistent)

1) Enterprises

- **Customer Relationship Management (CRM):** Databases store customer interactions, preferences, and sales data.
- **Supply Chain Management:** Databases track inventory levels, supplier information, and logistics.
- **Human Resources:** Manage employee data, payroll, and benefits using databases.
- **Financial Systems:** Handle accounting, budgeting, and financial reporting through database solutions.
- **Example: Salesforce CRM** is widely used by enterprises to manage customer relationships and sales data within a database.



2) Manufacturing

- **Production Tracking:** Databases monitor production schedules, inventory levels, and quality control.
- **Supply Chain Management:** Manage supplier information, procurement, and logistics using databases.
- **Product Lifecycle Management:** Track product design, development, and updates in a database.
- **Example: Toyota** uses databases to manage their Just-In-Time (JIT) manufacturing processes, ensuring efficient production and inventory management.



3) Banking and Finance

- **Account Management:** Databases store customer accounts, transactions, and balance information.
- **Loan Processing:** Manage loan applications, approvals, and repayments through a database system.
- **Fraud Detection:** Monitor transactions for unusual activity and potential fraud using database systems.
- **Example: JPMorgan Chase** uses advanced database systems to manage customer accounts, process transactions, and detect fraud.



4) Web-based Services

- **User Data Management:** Databases store user profiles, preferences, and activity logs.
- **Content Management:** Manage website content, including articles, images, and videos, using a database system.
- **Analytics:** Track user interactions, traffic patterns, and conversion rates through databases.
- **Example: Netflix** uses a sophisticated database system to manage user preferences, content recommendations, and streaming data.



5) Document Databases

- **Data Storage:** Document databases store semi-structured data in formats like JSON, BSON, or XML, allowing for flexible schema design and easy data retrieval.
- **Content Management:** Manage collections of documents such as articles, patents, research papers, and other unstructured content.
- **Scalability:** Document databases are designed to handle large volumes of data and can scale horizontally across multiple servers.
- **Use Cases:** Often used for content management systems, catalogues, and data lakes where data doesn't fit neatly into relational models.
- **Example: MongoDB** is a popular document database used by companies like **The New York Times** to manage and deliver vast amounts of news articles and media content.

The New York Times



6) Navigation Systems

- **Location Data:** Databases store geographic information including locations of places of interest, routes, and geographic features.
- **Real-Time Updates:** Manage real-time traffic data, route optimization, and point-of-interest information.
- **User Interaction:** Provide features like route planning, navigation guidance, and local searches.
- **Integration:** Often integrated with other systems such as weather services, local businesses, and emergency services to enhance user experience.
- **Example: Google Maps** uses a complex database system to manage and update location data, provide navigation services, and offer real-time traffic updates to millions of users worldwide.



There are two modes in which databases are used today

- **Online Transaction Processing (OLTP):**
 - Used by a large number of users for small data retrieval and updates
 - common in most database applications like banking, universities, and airlines.
- **Data Analytics/Online Analytical Processing(OLAP):**
 - Involves processing data to draw conclusions and create predictive models for business decisions.
 - Examples include loan approval, targeted advertisements, and manufacturing decisions.
 - Data mining combines AI and statistical techniques for efficient analysis of large databases.

- **A database system is a collection of interrelated data and a set of programs that allow users to access and modify these data.**
- A major purpose of a database system is to provide users with an abstract view of the data.
 - **Data models**
 - A collection of conceptual tools for describing data, data relationships, data semantics and constraints.
 - **Data abstraction**
 - Hide the complexity of data structures to represent data in the database from users through several levels of data abstraction.
- The structure of a database is defined by its data model, which includes tools to describe data, relationships, semantics, and consistency constraints.

Some questions to be explored??

- 1) List and describe three real-world applications of databases in different industries.
How do these applications benefit from using databases?
- 2) What are the two modes in which databases are used today?



Database Management Systems

Purpose of Database System

Anchor: Dr. Nagasundari S, Dept of CSE

Database Management Systems

Unit 1: Introduction to Database Management and SQL

Slides adapted from Author Slides of “Database System Concepts”, Silberschatz, H Korth and S Sudarshan, McGrawHill, 7th Edition, 2019. And Author Slides of Fundamentals of Database Systems”, Ramez Elamsri, Shamkant B Navathe, Pearson, 7th Edition, 2017.

Special Acknowledgement to all CSE Professors who contributed to this endeavor

Department of Computer Science and Engineering



File-based System:

- One way to keep information on a computer is to store it in permanent files.
- A company system has a number of application programs; each of them is designed to manipulate data files.
- These application programs have been written at the request of the users in the organization.
- New applications are added to the system as the need arises.

Let us take an example of a **University Organization system**. We need to store information about the following entities:

- Instructors, Students, Departments, Courses offered.
- One of the ways to keep this information is to store it in **operating file systems**
- To allow the users to manipulate the information, the system would have a number of application programs that manipulate the files including:
 - Addition of new students, instructors, and courses
 - Register students for different courses
 - Assign grades to students, compute GPA, and generate transcripts.

Scenario: The university decides to add a new major.

What specific steps must be taken to integrate a new major into the university's existing system?

1. Create a New Department:

- Set up a new department for the major.
- Create new permanent files for department information.

2. Develop New Application Programs:

- Write programs to manage rules for the new major.
- Ensure programs handle course registration, grading, and transcripts for the new major.

This typical file-processing system is supported by a conventional operating system. The system stores permanent records in various files, and it needs different application programs to extract records from and add records to, the appropriate files.

Consider the following situations:

- Double Major Students :

What if there is a student who has enrolled for a double major (Physics and Computer Science) how many times do we store all his details?

- Updating information :

What happens if a student changes their address or phone number? Is it guaranteed that the update reflects in all relevant files across departments?

- Efficient Data Retrieval:

The university transport department wants to find out all students who live within a particular postal code area. Can he quickly retrieve the data or does he have to build another application for it?

- Data Format Consistency:

What if files that store these data are written in different formats? Is it easy to access the data in this form?

- Adding New Constraints:

Suppose a new constraint is to be added, can it be done efficiently without having to make extensive changes to existing application programs?

There are a lot of **drawbacks to the file-processing system** :

- Data Redundancy and Inconsistency
- Difficulty in accessing the data
- Data isolation
- Integrity problems
- Atomicity problems
- Concurrent access anomalies
- Security problems

1. Data Redundancy & Inconsistency

Consider the question of the student enrolled for a double major. What would happen in this situation in case of file system?

Details of the students would be duplicated for each department - **Data Redundancy**

This would lead to the following problems:

- Takes up more storage space
- May also lead to **Data Inconsistency** in case of changes

The solution to this problem:

- Store information in centralized location thus Minimizes redundancy
- Changes can be made at a single spot, thus ensuring consistency
- This can be achieved through a database system

2. Difficulty in accessing the data

Consider that the university wants **to find all the students living in a specific postal code area**. How will we execute this in traditional file system?

- Either we need to get a list of all students and manually sift through it
- Or create another special program for this task

Both the above-mentioned are time-consuming and inconvenient

Thus we can see that Conventional file-processing systems are not designed for quick and efficient data retrieval, especially when new types of queries arise.

The solution to this problem:

- Central information store for fast retrieval
- Use a powerful query language for answering new queries
- Efficient ways to search and filter data for speeding up data retrieval
- This can be achieved through a database system

3. Data Isolation

Consider a concurrency situation, where identifying when and how changes made by one operation become visible to other concurrent users.

This is a challenge because New applications has to retrieve the appropriate data, which might be stored in various files.

A file-based system must manage, or prevent, concurrency by the application programs.

When an application opens a file, it will lock the file. This means that no one else has access to the file at the same time.

The solution to this problem:

- Ability to allow multiple transactions to access same data without interfering with each other
- Consistent view of the data

4. Integrity problems

Scenario: Each department has an **account for research funds, and the account balance must always be above 0.**

How can we implement this?

Each application program is created for maintaining departmental accounts but adding new constraints becomes challenging as it requires editing all the existing application programs.

The solution to this problem:

- Specify integrity constraints at the database level to specify rules directly in the schema
- System should enforce constraints automatically
- To add a new constraint, updating the schema alone will be sufficient, and the system will handle the rest automatically.
- This would be easily solved by using DBMS which provides all these facilities

5. Atomicity problems:

In a computer system failure can happen unexpectedly

Consider a banking system, a sum of 5000 rupees is getting transferred from account A to B.

What happens internally?

- An amount of 5000 Rs. gets deducted from account A
- The amount (5000 Rs.) gets credited to account B

Now **either both the steps should be performed or none should be executed.**

In a file system, The program might successfully deduct 5000 from account A but fail to credit it to account B.

This results in an **inconsistent** database state, with money missing from one account and not added to the other.

The solution to this problem:

- Atomicity should be ensured
- Automatic rollback in case of failure
- DBMS would ensure this and makes it reliable and consistent even if failure occurs

6. Concurrent Access Anomalies

Consider a bank transaction example. There is a bank account (Account A) with a balance of \$10,000.

- Two bank clerks attempt to deduct money from Account A at almost the same time - one deducts \$500 and the other \$100.
- Since they work concurrently, they both read the initial balance of \$10,000, subtract their respective amounts, and write back the results.
- Depending on which one writes the new balance last, the account could end up with either \$9,500 or \$9,900 instead of the correct value of \$9,400.

The solution to this problem:

- Transaction and Locking to be implemented
- Transactions ensure atomicity
- Locking prevents conflicting changes.
- The DBMS would provide these and maintains data consistency and integrity

7. Security problems

Imagine a university using a file-based system. Over time, they've added various application programs without a central security system.

As a result, payroll personnel might accidentally see and even change academic records, compromising data privacy and confidentiality because everyone has the same permissions.

The solution to this problem:

- Create a provision for administrator to define access controls and permissions for each user or group.
- These permissions should be enforced at the database level, regardless of the application programs used.
- DBMS supports various methods for introducing security like User Authentication, Data Encryption, Auditing and Logging, Centralized security management

- **Self-describing nature of a database system**
 - The database system contains not only the database itself but also a complete definition or description of the database structure and constraints (Metadata).
- **Insulation between programs and data, and data abstraction**
 - The structure of data files is stored in the DBMS catalog separately from the access programs (program-data independence)
- **Support multiple views of the data**
 - DBMSs support different types of users, each of whom may require a different perspective or view of the database. A view may be a subset of the database or it may contain virtual data that is derived from the database files but is not explicitly stored.
- **Sharing of data and multi user transaction processing**
 - A multi user DBMS, allows multiple users to access the database at the same time. It ensures that data changes are effected in a controlled manner so that the result of the changes are correct

- Controlling redundancy in data storage.
- Sharing of data among multiple users.
- Restricting unauthorized access to data
- Providing persistent storage for program Objects
 - E.g., Object-oriented DBMSs make program objects persistent
- Providing Storage Structures and search techniques for efficient Query Processing
- Providing backup and recovery
- Providing multiple interfaces to different classes of users.

- Representing complex relationships among data.
- Enforcing integrity constraints on the database.
- Drawing inferences and actions from the stored data using deductive and active rules and triggers
- Potential for enforcing standards
- Reduced application development time
- Flexibility to change data structures
- Availability of current information
- Economies of scale

Identify the Problem: Case Study Applications

1. Scenario 1: Banking Transactions:

At a bank, two customers attempt to withdraw money from the same account simultaneously. Due to a lack of concurrency control, the system fails to manage the simultaneous transactions correctly, resulting in the account being overdrawn.

Problem Type: Concurrent Access Anomalies



Scenario 2: Customer Records:

A company stores customer information in multiple spreadsheets, with each department maintaining its own copy. Occasionally, customer addresses and contact details differ between spreadsheets, leading to confusion and errors in communication.

Problem Type: Data Redundancy and Inconsistency

1

Name	Address	Phone
John Doe	123 Elm Street	555-9067
Grace	12 Oak Street	123-8907
Judy	45 York Lane	237-1123

2

Name	Address	Phone
Grace	12 Oak Street	123-8907
John Doe	129 Blueberry Lane	555-9067
Judy	45 York Lane	237-1123

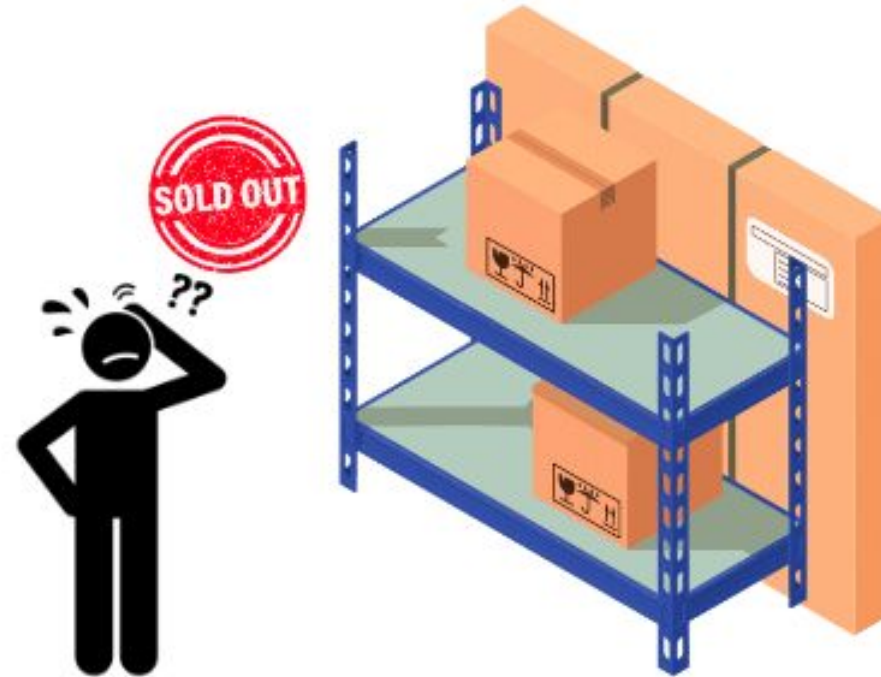
3

Name	Address	Phone
John Doe	123 Elm Street	555-9067
Judy	45 York Lane	237-1123
Grace	12 Oak Street	123-8907

Scenario 3: Inventory and Orders:

An online shopping system has two transactions running at the same time. Transaction A is updating the stock quantity of a product from **10 to 5**, but the transaction has not yet been committed. At the same time, Transaction B reads the stock quantity and sees **5**.

Problem Type: Data Isolation

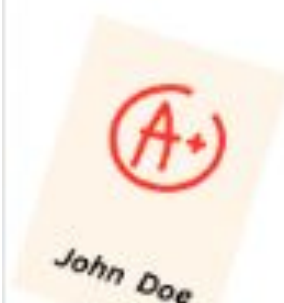


Scenario 4: Student Grades:

A university records student grades manually, and sometimes errors are made in entering the data, such as typing grades incorrectly. There is no system in place to check for or correct these errors automatically.

But, if grade is entered as 'x' which is not a valid grade then constraints can be added to handle.

Problem Type: Integrity Problems



Name	Grade
John Doe	X
Judy	A
Grace	C

Some online resources -

- <https://dev.mysql.com/doc/>
- <https://www.oracle.com/in/database/what-is-database/>
- https://docs.oracle.com/cd/E17952_01/index.html
- <https://downloads.mysql.com/docs/mysql-tutorial-excerpt-8.0-en.a4.pdf>



THANK YOU

Anchor: Dr. Nagasundari S, Dept of CSE

nagasundaris@pes.edu